ARAM SALEHI

DATA SCIENTIST | APPLIED MACHINE LEARNING & COMPUTER VISION

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PROFILE

Applied ML and medical-image analysis specialist with completed PhD research at Radboudumc and LUMC (defense pending). Develops end-to-end Python/PyTorch imaging pipelines, from dataset preparation and image segmentation to model benchmarking, quality control and reproducible inference. Experienced in low-data imaging settings, cross-domain adaptation and collaborating with clinical and engineering teams. Interested in transferring these skills to pharmaceutical machine-vision inspection and defect-detection systems.

TECHNICAL SKILLS

Computer vision & ML: Python, PyTorch, TensorFlow, OpenCV, scikit-learn; segmentation (U-Net, Swin-UNETR), image enhancement, reconstruction, domain adaptation, quantitative image analysis.

Data & model validation: NumPy, Pandas, SQL; image preprocessing, dataset quality control, held-out testing, baseline comparisons, error analysis, ICC, agreement/repeatability evaluation.

Engineering & deployment: Git/GitHub, Docker, Linux, NVIDIA GPU/HPC workflows, inference pipelines, experiment tracking (Weights & Biases).

Agentic AI & RAG: LangChain, OpenAI Python SDK (tool/function calling), Chroma vector database, embeddings, retrieval-augmented generation, Streamlit, Ollama, Hugging Face; hands-on prototype development.

PROFESSIONAL EXPERIENCE

PhD Researcher — Machine Learning & Medical Imaging

Radboud University Medical Center / Leiden University Medical Center | Nov 2022 – Oct 2026

- Built an end-to-end automated MRI segmentation and area-measurement workflow incorporating preprocessing, ROI localisation, inference and QC; assessed repeatability across 2,690 scans from 899 participants (ICC 0.933).
- Benchmarked transformer-based segmentation against alternative 3D architectures using consistent train/test splits, quantitative metrics and systematic failure review; achieved mean selected-slice Dice of 0.858 on 47 held-out scans.
- Developed cross-field MRI image-translation models supporting five field strengths and three contrasts, addressing dataset heterogeneity and robustness to changing acquisition conditions; contributed to a top-three MRIxFields 2026 team.
- Designed and evaluated 3D enhancement/denoising models for low-field MRI in limited-data settings; placed third in the MICCAI 2025 ultra-low-field MRI enhancement challenge.
- Packaged inference workflows in Docker, trained models on NVIDIA GPUs and Snellius HPC, tracked experiments using Weights & Biases, and communicated validation findings with multidisciplinary collaborators.

Researcher — Deep Learning (part-time)

University of Isfahan | 2019 – 2022

- Developed TensorFlow U-Net pipelines for six-organ CT segmentation, MRI-to-CT synthesis and neural-network-based CT reconstruction; conducted preprocessing and quantitative evaluation in limited-data imaging tasks.

SELECTED PROJECTS — RELEVANT TO LUO

Automated Anterior Commissure Segmentation & QC

Medical imaging model development | PyTorch, Swin-UNETR, FreeSurfer

Integrated ROI extraction, supervised segmentation, automated measurements and QC checks in a reproducible pipeline. Compared models against baselines and reviewed failed preprocessing and ambiguous segmentation outputs. Demonstrates the offline qualification and dataset-quality focus relevant to image-based pharmaceutical inspection.

Cross-Domain MRI Harmonisation

Model adaptation | PyTorch, conditional Swin-UNETR/CUT, Docker

Implemented a unified conditional model for image translation across multiple MRI field strengths and contrasts. Applied simulated-data pretraining, paired fine-tuning and quantitative validation to improve robustness across imaging conditions.

RAG Chatbots & Tool-Calling AI Agents

Personal portfolio projects | LangChain, OpenAI SDK, Chroma, Streamlit, Ollama

Built document-grounded Q&A chatbot prototypes using chunking, embeddings, Chroma retrieval and LangChain; explored relevance scoring and deployed a Streamlit interface. Practised agent/tool workflows with the OpenAI Python SDK, including function-calling interfaces and Python tool execution for structured tasks. These are prototype projects, not production pharmaceutical applications.

EDUCATION

PhD research in AI & Data Science for Medical Imaging | 2022–2026

Radboudumc / LUMC | Research completed; doctoral defense pending

Thesis: Methodological Advances to Promote Accessible Neuroimaging

MSc, Radiation Application | University of Isfahan | 2016–2018

BSc, Electronic Engineering (Telecommunications) | Urmia University | 2010–2014

SELECTED PUBLICATIONS & RECOGNITION

- Third place — MICCAI 2025 Ultra-Low-Field MRI Enhancement Challenge (ULF-EnC).
- Top-three team — MRIxFields 2026; posters on cross-field MRI harmonisation.
- Salehi et al., “Denoising low-field MR images with a deep learning algorithm based on simulated data,” Journal of Magnetic Resonance (2025).
- Co-author, “The potential of low-field MRI for global dementia care,” Nature Reviews Neurology (2026).

ADDITIONAL TRAINING & LANGUAGES

Applied Data Science Lab (WorldQuant University); deep learning and TensorFlow advanced techniques; SQL/Snowflake training.

English C1 (IELTS 7) | Dutch A2 (actively learning) | Persian advanced | Kurdish native